I made all my tables by copying my python results into Google Gemini and telling it to make latex tables from the output.

1 Problem 0 – Provided Code

1.1 Deriving an Analytical Expression

Assume the current neutron population is related to the initial population like some exponential decay.

$$N = N_0 \exp(-\lambda) \tag{1}$$

The input for our relationship is the current distance from the initial location, x, which we assume is proportional to λ because that is the only free variable.

$$x \propto \lambda \tag{2}$$

We expect each material to have some quantity describing neutron interacts inside the material. We also know λ is dimensionless, so we need to multiply x by some quantity with a unit of [1 / length]. We will call the quantity describing neutron interaction "cross section," Σ, and assume it can be expressed with units of [1 / length].

$$[x\Sigma] = 1 \tag{3}$$

From a unit analysis, we can see λ has the same units as $x\Sigma$. Also, N/N_0 is the fraction of unabsorbed neutrons. So, we arrive at the final expression:

$$p_{abs} = \exp(-x\Sigma) = \exp(-1.0 \cdot 5.0) = 0.00673795 \tag{4}$$

1.2 Purely Absorbing Shield

In the previous part, the analytical expression for the transmission probability agrees very well with the code seeing as all trials, except the one with 10,000 samples, agree with the analytical result to at least the first significant figure. Also, we see the probability tend to the analytical expression as the number of samples increases.

Table 1: Purely Absorbing Shield Results

$N_{\mathbf{samples}}$	$P_{\mathbf{ref}}$	$R_{\mathbf{ref}}$	$\mathbf{FOM_{ref}}$	$P_{\mathbf{abs}}$	$R_{\mathbf{abs}}$	$\mathbf{FOM_{abs}}$	$P_{\mathbf{trn}}$	$R_{\mathbf{trn}}$	$\mathbf{FOM_{trn}}$
10,000	0.0	1.0	67.5237	0.993000	0.000840	9.5787×10^7	0.007000	0.119104	4759.98
50,000	0.0	1.0	13.7293	0.993740	0.000355	1.0897×10^8	0.006260	0.056346	4324.34
100,000	0.0	1.0	5.8328	0.993450	0.000257	8.8467×10^7	0.006550	0.038945	3845.68
500,000	0.0	1.0	1.3577	0.993368	0.000116	1.0168×10^8	0.006632	0.017308	4532.22
1,000,000	0.0	1.0	0.6540	0.993246	0.000082	9.6176×10^7	0.006754	0.012127	4447.08

Table 2: Shield with 10% Isotropic Scattering Results

$N_{\mathbf{samples}}$	$P_{\mathbf{ref}}$	$R_{\mathbf{ref}}$	$\mathbf{FOM_{ref}}$	$P_{\mathbf{abs}}$	$R_{\mathbf{abs}}$	$\mathbf{FOM_{abs}}$	$P_{\mathbf{trn}}$	$R_{\mathbf{trn}}$	$\mathbf{FOM_{trn}}$
10,000	0.016000	0.078422	7637.87	0.976800	0.001541	1.9777×10^7	0.007200	0.117426	3406.58
50,000	0.016720	0.034295	7779.90	0.976140	0.000699	1.8718×10^7	0.007140	0.052736	3290.22
100,000	0.016210	0.024635	7821.41	0.976550	0.000490	1.9768×10^7	0.007240	0.037030	3461.77
500,000	0.016424	0.010944	8120.41	0.976382	0.000220	2.0104×10^7	0.007194	0.016614	3523.81
1,000,000	0.016570	0.007704	8114.72	0.976057	0.000157	1.9633×10^7	0.007373	0.011603	3577.28

1.3 Shield with 10% Isotropic Scattering

To examine these results, we will use the relative uncertainty and the figure of merit (FOM).

$$R_{\bar{x}} \propto \frac{\sigma_x}{\sqrt{N}} \approx \frac{C_1}{\sqrt{N}} \tag{5}$$

$$FOM \propto \frac{1}{\left(\frac{C_1}{\sqrt{N}}\right)^2 (C_2 N)} \tag{6}$$

If our simulation is reliable and valid, we need to make ensure the central limit theorem applies. For all cases, here is the relative uncertainty times the square root of the number of particles. From Table 3, we observe our quantity of interest is roughly constant for all sample numbers for both cases.

Table 3: Central Limit Theorem Validation

	Case 1 (Transmission)		**Case 2 (Transmission)**	
$N_{\mathbf{samples}}$	$R_{\mathbf{trn}}$	$R_{\mathbf{trn}} \times \sqrt{N}$	$R_{\mathbf{trn}}$	$R_{\mathbf{trn}} \times \sqrt{N}$
10,000	0.119104	**11.9104**	0.117426	**11.7426**
50,000	0.056346	**12.5994**	0.052736	**11.7921**
100,000	0.038945	**12.3151**	0.037030	**11.7101**
500,000	0.017308	**12.2386**	0.016614	**11.7479**
1,000,000	0.012127	**12.1270**	0.011603	**11.6030**

We also see the FOM is oscillating about some constant as the number of sampled particles changes. This is great as a roughly constant FOM is required for a reliable MC simulation.

2 Problem 1 – Multi-Region

Table 4: Multi-Region Results

$N_{\mathbf{samples}}$	$P_{\mathbf{ref}}$	$R_{\mathbf{ref}}$	$\mathbf{FOM}_{\mathbf{ref}}$	$P_{\mathbf{abs}}$	$R_{\mathbf{abs}}$	$\mathbf{FOM}_{\mathbf{abs}}$	$P_{\mathbf{trn}}$	$R_{\mathbf{trn}}$	$\mathbf{FOM}_{\mathbf{trn}}$
10,000	0.529600	0.009425	66220.52	0.466600	0.010692	51452.16	0.003700	0.164095	218.44
50,000	0.531420	0.004199	70667.50	0.464140	0.004805	53971.35	0.004420	0.067118	276.64
100,000	0.531390	0.002970	69248.38	0.464540	0.003395	52979.02	0.004060	0.049528	248.94
500,000	0.529660	0.001333	68906.41	0.466618	0.001512	53530.03	0.003720	0.023144	228.47
1,000,000	0.531037	0.000940	69738.92	0.465373	0.001072	53609.20	0.003589	0.016662	221.83

From Table 4, we see the probability of reflection is roughly the same across all trials. In contrast, the associated relative uncertainty decreases as the number of trials decreases as we expect from our definition of relative uncertainty, which can be seen in Table 5. We notice the exact same trend for both transmission and absorption.

Table 5: Central Limit Theorem Validation for Multi-region Case

$N_{\mathbf{samples}}$	$R_{\mathbf{trn}}$	$R_{\mathbf{trn}} \times \sqrt{N}$
10,000	0.164095	16.4095
50,000	0.067118	15.0080
100,000	0.049528	15.6621
500,000	0.023144	16.3653
1,000,000	0.016662	16.6620

3 Problem 2 – Incident Angle

I will note the probability percent values are given to the fourth decimal place. If the simulation runs with the maximum of 1e6 particles, a 0.0001% transmission probability means one of the million particles made it to the location.

Table 6: 0.2 Scattering Ratio

θ	**# of histories**	**Transmission (%)**	$R_{\mathbf{trans}}$	**Reflection (%)**	$R_{\mathbf{ref}}$	**Absorption (%)**	$R_{\mathbf{abs}}$
0°	1,000,000	0.0061%	0.12803	3.5471%	0.00521	96.4468%	0.00019
30°	1,000,000	0.0014%	0.26726	3.8170%	0.00502	96.1816%	0.00020
60°	1,000,000	0.0002%	0.70711	5.0744%	0.00433	94.9254%	0.00023

Table 7: 0.8 Scattering Ratio

θ	**# of histories**	**Transmission (%)**	$R_{\mathbf{trans}}$	**Reflection (%)**	$R_{\mathbf{ref}}$	**Absorption (%)**	$R_{\mathbf{abs}}$
0°	92,343	0.1083%	0.09995	28.4602%	0.00522	71.4315%	0.00208
30°	133,912	0.0747%	0.09996	30.3408%	0.00414	69.5845%	0.00181
60°	269,612	0.0371%	0.09998	36.7513%	0.00253	63.2116%	0.00147

4 Problem 3 – Implicit Capture with Russian Roulette

For the first case, the percentages between problem 3 and problem 2 are pretty close to the same for all three: transmission, reflection, and absorption. However, the relative uncertainties were globally 2 to 3 times lower when implicit capture was used.

Table 8: 0.2 Scattering Ratio with Implicit Capture

θ	# of histories	Transmission (%)	R_{trans}	Reflection (%)	R_{ref}	Absorption (%)	R_{abs}
0°	1,000,000	0.0048%	0.12780	3.5275%	0.00202	96.4677%	0.00007
30°	1,000,000	0.0013%	0.22402	3.8485%	0.00191	96.1502%	0.00008
60°	1,000,000	0.0002%	0.28411	5.0819%	0.00160	94.9180%	0.00009

Unlike the first case where the maximum number of iterations was reached, the second case converges because the convergence criteria is met. Similar to the first case, the probabilities are roughly the same, but the same convergence criteria was achieved in 3, 6, and 10 times fewer particle histories for the 0°, 30°, and 60° cases, respectively. From Implicit capture, we see a massive performance boost, especially for the angled problems where vanilla MC is far outshone by implicit capture.

Table 9: 0.8 Scattering Ratio with Implicit Capture

θ	# of histories	Transmission (%)	R_{trans}	Reflection (%)	R_{ref}	Absorption (%)	R_{abs}
0°	22,308	0.1321%	0.09988	28.5560%	0.00718	71.3120%	0.00287
30°	23,864	0.0981%	0.09981	30.0706%	0.00663	69.8314%	0.00285
60°	26,738	0.0412%	0.09986	37.2492%	0.00521	62.7096%	0.00309

5 Problem 4 – Geometric Splitting with Russian Roulette

In both problem 2 and problem 3, the maximum number of iterations was reached for every angle. However, with geometric splitting, we reached the convergence criteria for the 0° and 30° cases. However, the 60° case did not converge. We also see the relative uncertainties of reflection and absorption probability are higher in the 0° and 30° cases with geometric splitting. The reason these uncertainties increased is because geometric splitting, when set up properly, efficiently steals samples from low variance regions and "gives" these samples to high variance regions. The 60° probabilities did not change because the maximum number of iterations was reached.

Table 10: 0.2 Scattering Ratio with Geometric Splitting

θ	**# of histories**	**Transmission (%)**	$R_{\mathbf{trans}}$	**Reflection (%)**	$R_{\mathbf{ref}}$	**Absorption (%)**	$R_{\mathbf{abs}}$
0°	129,667	0.0057%	0.09952	3.4110%	0.01478	96.5761%	0.00054
30°	393,745	0.0018%	0.09951	3.8847%	0.00793	96.1212%	0.00033
60°	1,000,000	0.0002%	0.20621	5.0795%	0.00432	94.9186%	0.00023

For the second case, the probabilities are roughly the same as in problem 2. However, convergence was achieved with an even lower number of samples for every angle than both problem 2 and problem 3. We also notice the same pattern as in the previous case – the relative uncertainties on absorption and reflection are higher than in problem 2. Stealing particles from areas of low variance and moving them to areas of high variance can dramatically reduce the number of histories required for a MC simulation.

Table 11: 0.8 Scattering Ratio with Geometric Splitting

θ	**# of histories**	**Transmission (%)**	$R_{\mathbf{trans}}$	**Reflection (%)**	$R_{\mathbf{ref}}$	**Absorption (%)**	$R_{\mathbf{abs}}$
0°	7,799	0.1122%	0.09988	29.5422%	0.01752	70.4890%	0.00800
30°	11,717	0.0789%	0.09980	30.9550%	0.01382	68.9106%	0.00664
60°	22,648	0.0447%	0.09970	37.1865%	0.00865	62.8414%	0.00531